

### ABSTRACT

In the Study region of El Paso county Colorado, processed data has been used to answer this question: Can Land Surface Temperature (LST) and Normalized Difference Vegetation Index (NDVI) have an effect and correlation to the monthly average electricity bill by Census Tract in El Paso County, CO? The study finds that the average monthly electricity bill has more of a correlation to NDVI than it does LST. This shows a slight correlation that more greenery around your house's location can overall lower your electricity bill. Meanwhile, we see a strong correlation between LST and NDVI, stating that the more greenery in a region, the lower the surface temperature.

### INTRODUCTION

NDVI is a measurement of the density and health of the current vegetation cover in a small region. According to Grogani et al.(2013) and **Figure 1,2, and 7.** , LST has a strong correlation of dropping when NDVI is increased (especially in an urban environment). This is because vegetation can absorb solar radiation, cooling down the regions around it. During the summers, it is expected that El Paso County residences use their Central Air Conditioning (AC) more often, thus raising the overall electricity bill. Can NDVI and LST have an influence on how much electricity you need to use?

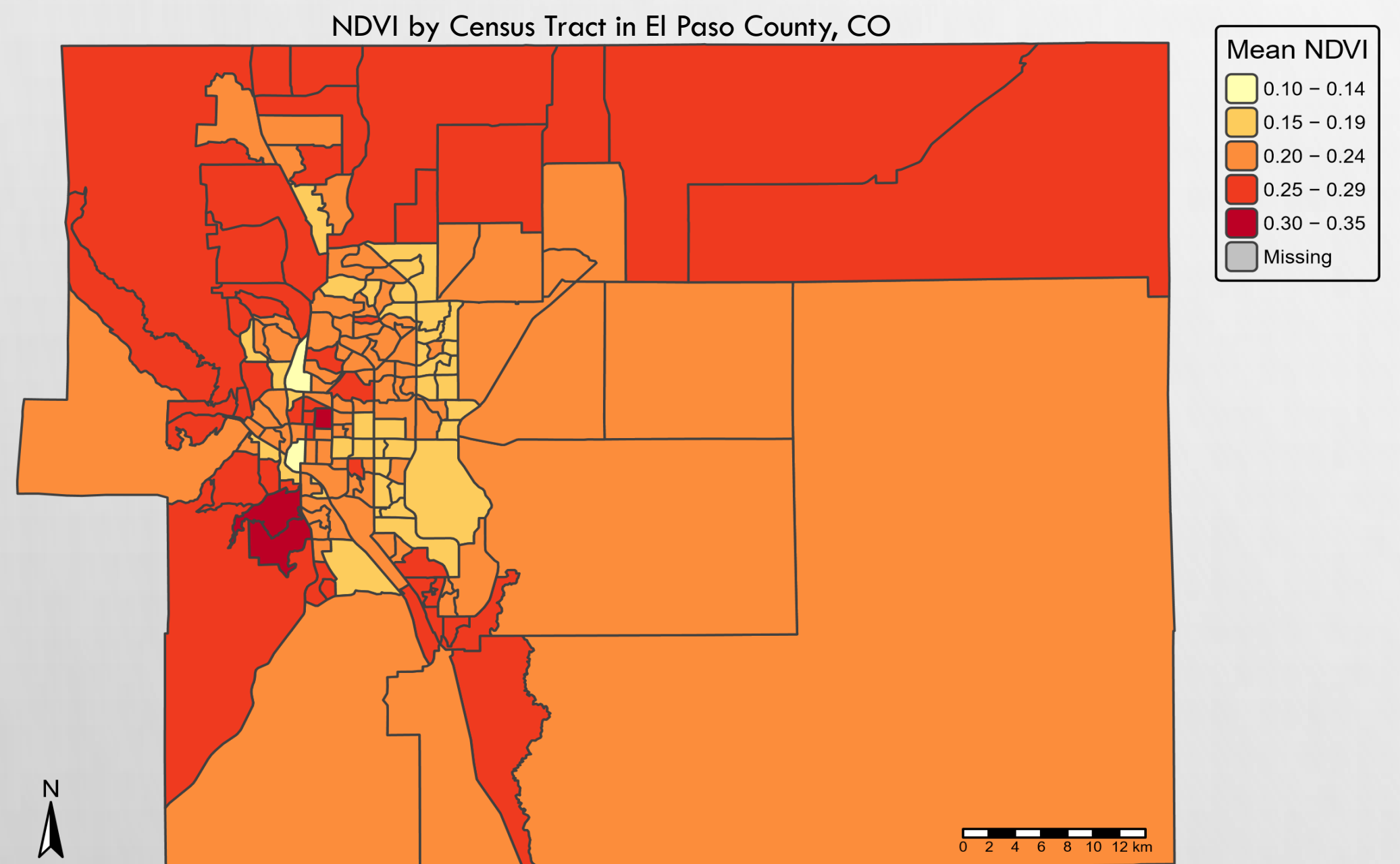


Figure 1. NDVI by Census Tract (Polaski, 2025)

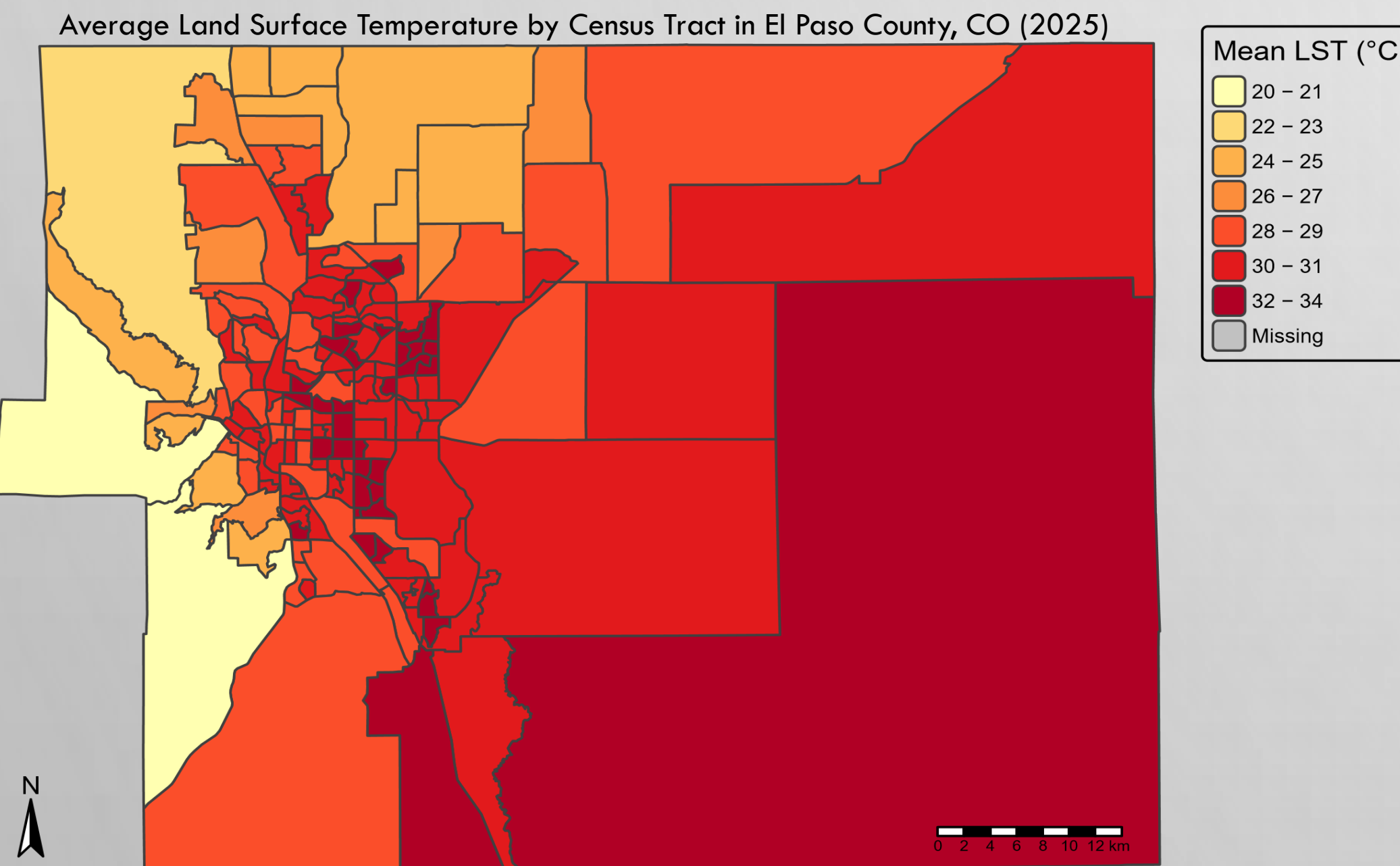


Figure 2. LST by Census Tract (Polaski, 2025)

### METHODOLOGY

**Study Area:** El Paso County Colorado, home to places like Monument, Manitou Springs, and Colorado Springs. This region was selected based on personal interest with El Paso County but also selected for its urban structures and natural spaces. This study area shows great potential to understand the topic of LST, NDVI, and monthly electricity bills.

#### Data collection:

- Shapefile: El Paso county shapefile (downloaded from Lab 3)
- Census API key: U.S. Census website was not working at the time, thus the usage of an API key.
- LANDSAT 8 Raster data: USGS.gov website (Imagery bundle), but only using TIFF bands 3, 4, 10 and MTL text file.
- R Studio: R studio was used to process and visualize data.
- Chat GPT: To help create a conversion code to use Band 10 in order to turn LANDSAT 8 units from Kelvin to Celsius.

#### Data processing:

Using the Census API key, I was able to create a tm\_shape with records from the 2022 Census variables. I extracted all electricity variables from 2022 Census tracts and used those variables to put into a map format from in class Labs. For the LST raster, I had to combine the El Paso county shape file and Census tract neighborhoods into the Band 10 satellite imagery. I used this shapefile to “crop” Band 10 into just the study area. The units for Band 10 had to have been converted and that is when Chat GPT created an equation for my MTL textfile (associated with all LANDSAT 8 bands) as seen in **Figure 7.** NDVI imagery was created with bands 3 and 4 using rast() in replacement of st\_read() or csv.read() for naming the bands. Doing this allowed the NDVI equation:  $ndvi <- (nir - red) / (nir + red)$  to be calculated. NDVI mean and LST mean was extracted from each Census tract using the API key and tracts value. AC data was imported into the tracts value to create a map expressing the R values between all the varibales. Maps and plots were created using the function (“tmap”) and (“ggplot2”). By visually seeing the maps and comparing the data, it was easy to see their relationships with the R value.

### RESULTS

The study found that LST and NDVI have a R value of -0.56, showing that the correlation is negative. As NDVI increases, LST decreases. LST and monthly average electricity cost show an R value of 0.05. This positive correlation shows the slight influence LST has on the average monthly electricity bill. As LST increases, so does the electricity bill. However, the most interesting correlation was the relationship between NDVI and monthly electricity with an R value of -0.07. This means vegetation has more of an effect on the average monthly electricity bill than land surface temperature. The negative correlation shows that as vegetation cover/vegetation health increases in a census tract, the average monthly electricity bill will decrease. This is likely due to vegetation correlation to LST and the small influence LST has on electricity bills. NDVI can also provide more shade for residents, influencing their overall monthly usage of AC.

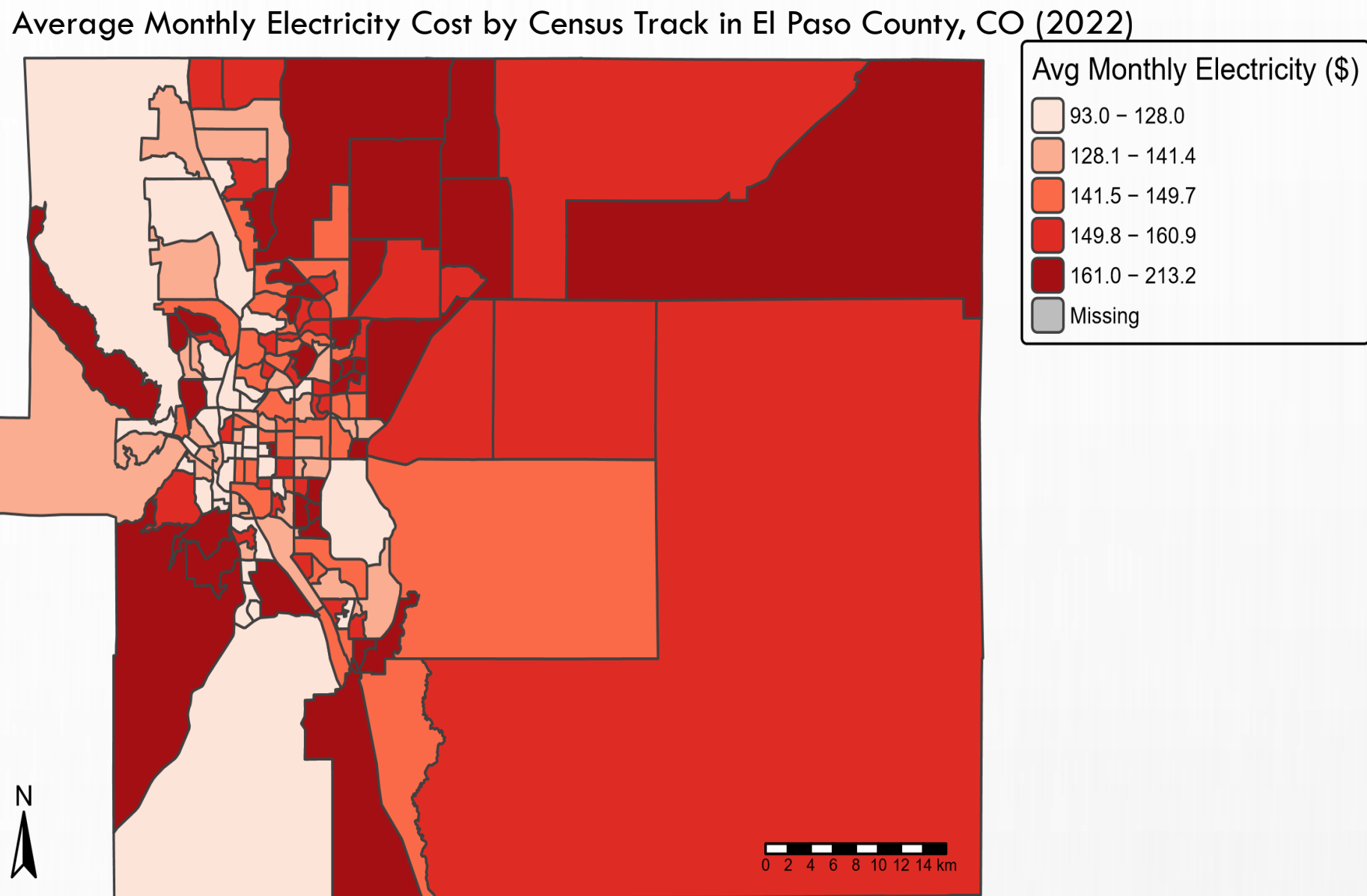


Figure 3. Average Electricity Cost by Census Tract (Polaski, 2025)

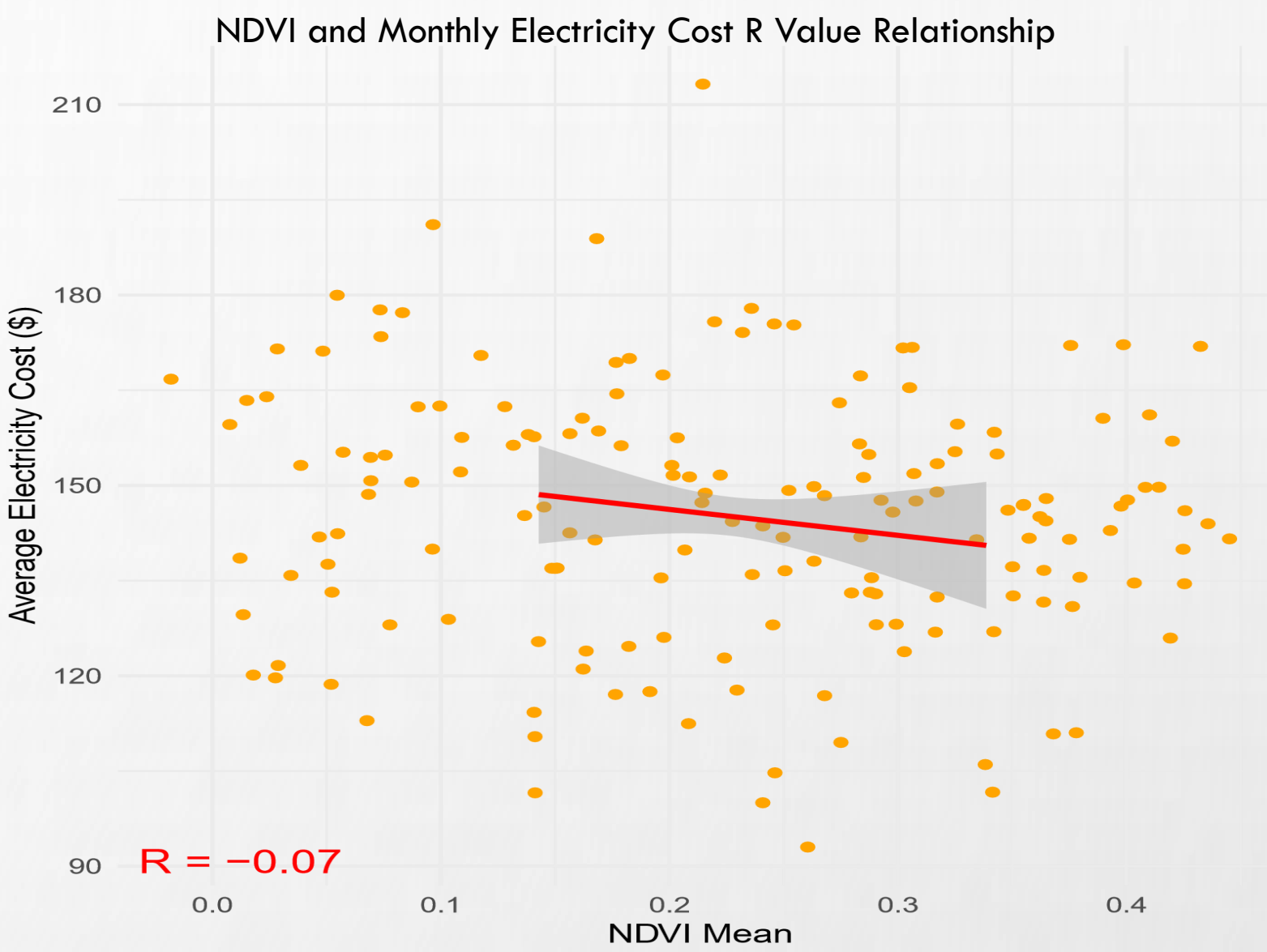


Figure 4. NDVI and Electricity R Value (Polaski, 2025)

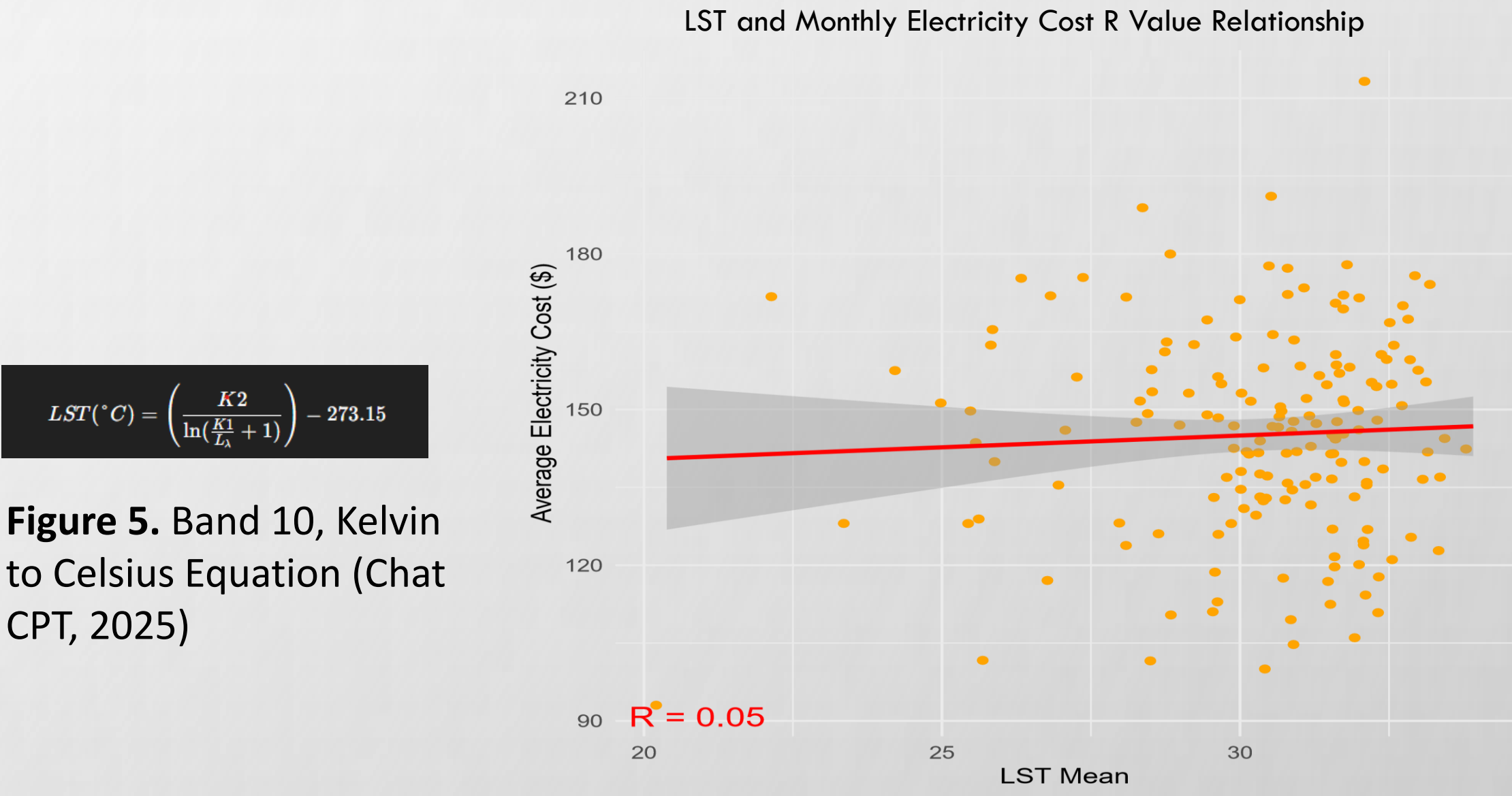


Figure 6. LST and Electricity R Value (Polaski, 2025)

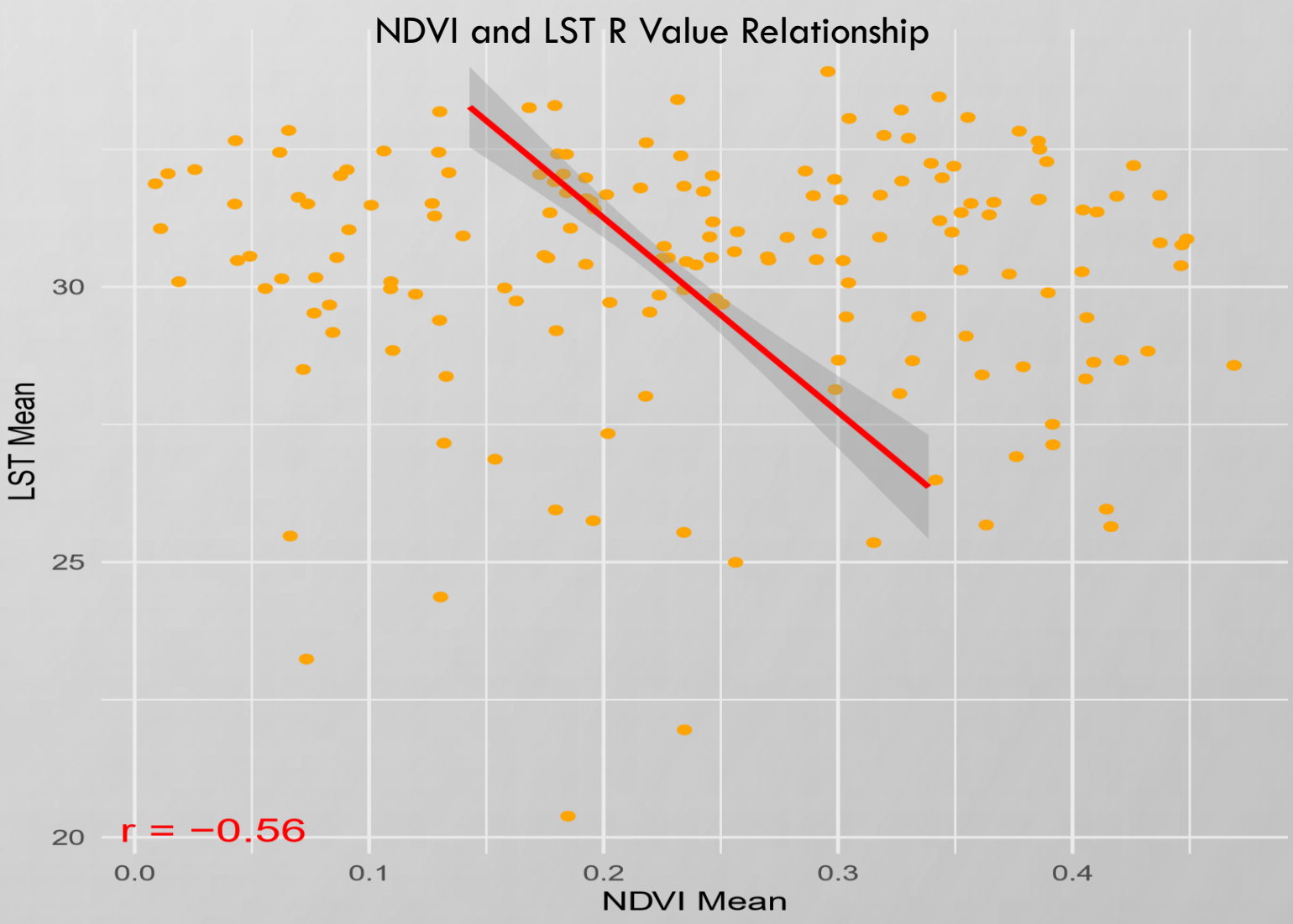


Figure 7. NDVI and LST R Value Relationship (Polaski, 2025)

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### Data Source

1. Grogani et al.(2013). *The Relationship between NDVI and LST in the Urban area of Mashhad, Iran, Volume (6), page 5.* [Tabriz-E-libre.pdf](#)
2. N.n. (2022)United States Census Bureau(2022) [Census.gov | U.S. Census Bureau Homepage](#)
3. U.S. Geological Survey. (2025). *Landsat 8 OLI/TIRS Collection 2 Level-2 science products* [June 17, 2025]. U.S. Department of the Interior. <https://earthexplorer.usgs.gov/>